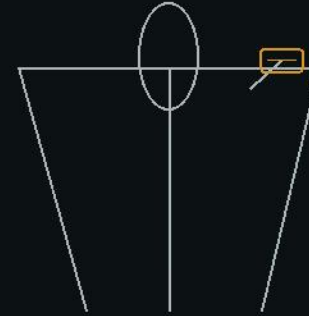


CREW-SHARABLE SITUATIONAL AWARENESS · CIVILIAN COMPUTER VISION

EAGLE EYE

Shared Detection for Phone Crews

Live Shared Detection for Phone Crews



Civilian computer-vision prototype

Headline

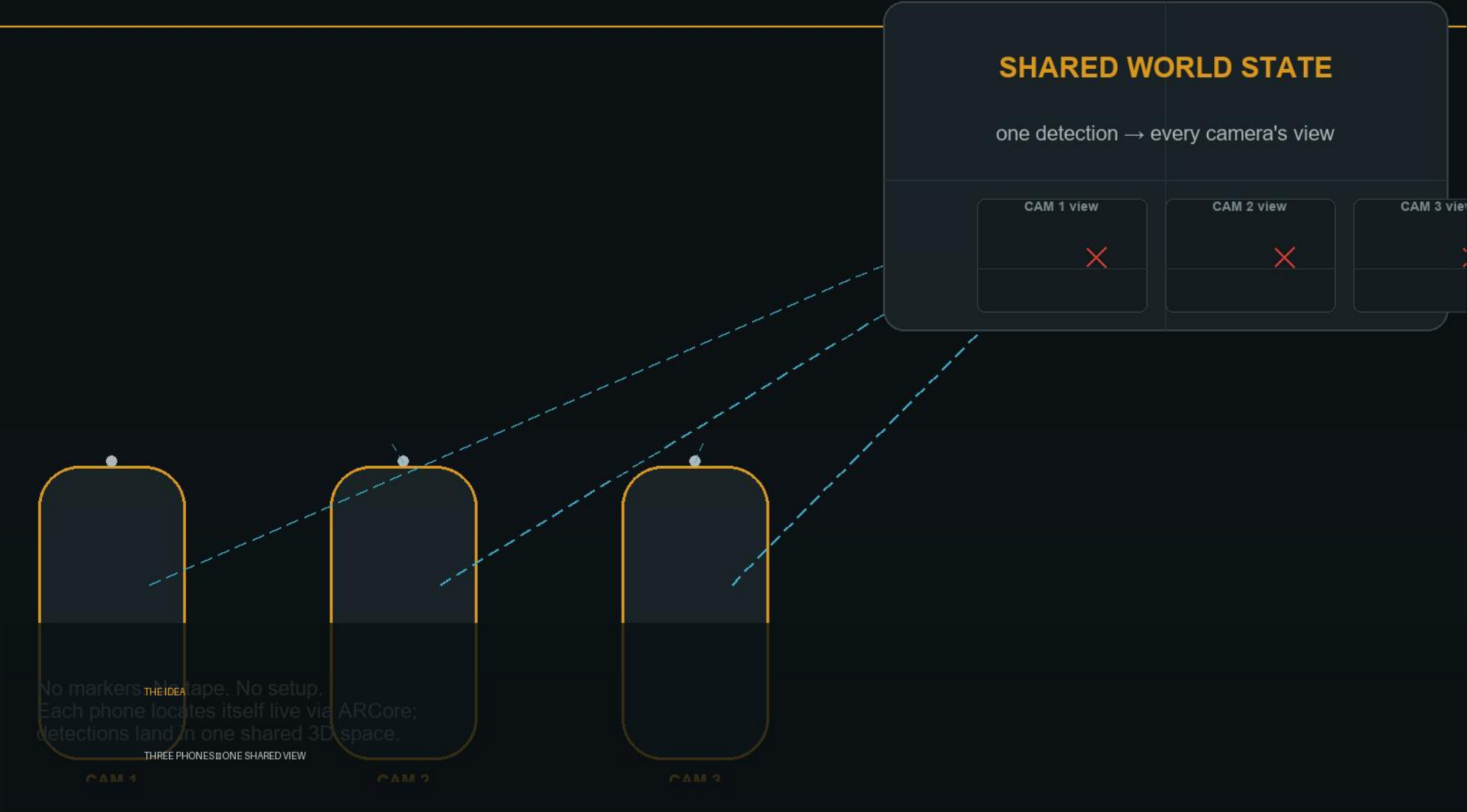
WebXR + WebRTC + YOLOv8rZero configuration — works in any room

Tech

Promise

WebXR + WebRTC + YOLOv8n · Zero configuration  works in any room

THREE PHONES — ONE SHARED VIEW



Each phone streams its view and its live ARCore location to one server. When a person is spotted, every other screen shows a red X at their real 3D position (even through a wall).

HOW IT WORKS — THREE STREAMS INTO ONE SHARED WORLD



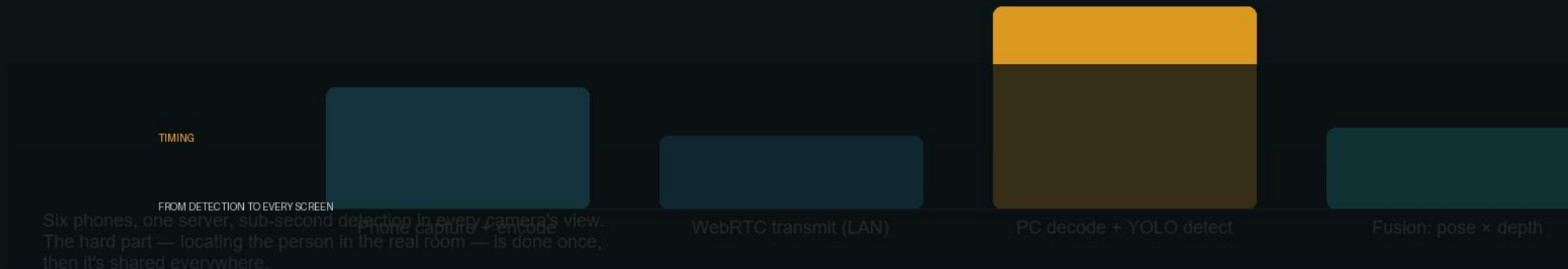
Real upside: all streams are plain HTTPS / WebRTC — nothing exotic. A phone trusts the cert (one tap), enters AR, and the rest is automatic.

HOW IT WORKS

Three plain streams (ARCore pose, camera frame, video) converge in a fusion engine that turns each detection into one real 3D point, then broadcasts it to every viewer. Nothing exotic: HTTPS, WebRTC, a WebSocket. A phone trusts the cert (one tap), enters AR, and it just works.

$\leq 1.5 \text{ s}$

end-to-end: detected $\rightarrow$ every viewer



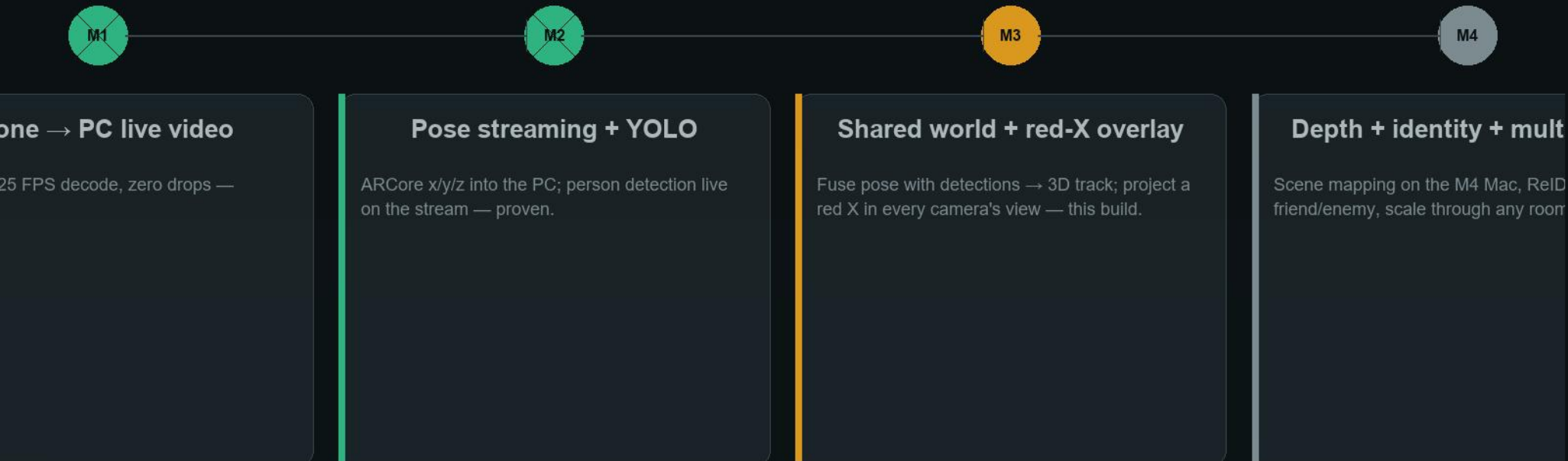
1.5 seconds, end-to-end captured, decoded, detected, fused, and broadcast to every viewer. The hard part locating the person in the real room is done once, then shared everywhere.

STATUS TODAY

☐	Phone → PC live video (WebRTC) ~25 FPS decode, zero drops	M1
☐	ARCore pose streaming to the PC ARCore x/y/z + quaternion, two phones simultaneously	M2
☐	YOLOv8n person detection conf 0.35, live on the stream, ~15 FPS on CPU	M2
☐	Clock sync + latency telemetry NTP-style sync; rtt and one-way latency every 5 s	M2
○	Fusion: detections × pose → 3D track depth from bbox height; one world per person	M3
○	Cross-camera red-X overlay project a person's point into every other camera's view	M3
○	Shared world state broadcast /world JSON → live viewer page	M3
○	Person identity across cameras v1 nearest-ray matching; ReID later	M4
○	Friend vs enemy distinction v1 marks all people	M4
○	Scene mapping / walls in 3D depth-everything on the M4 later	M4

M1–M2 proven · M3 in progress · M4 planned

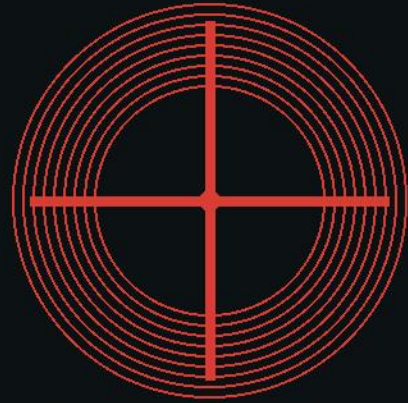
ROADMAP



WHATS NEXT

Today, we have M1 + M2 proven and M3 is the current build.
M4 makes the prototype serious; M3 makes it useful.

Today: M1 and M2 are proven, M3 is the current build. M4 makes the prototype serious: scene mapping, identity across cameras, friend vs enemy. But M3 is what makes it useful right now.



ONE PERSON IN EVERY VIEW

Eagle Eye — shared situational awareness for phone crews

Civilian computer-vision prototype